

Appendix A

Rolling-Horizon UC with RoCoF Constraints

Rolling-Horizon Setting

The annual rolling-horizon UC model covers 8760 operating hours. Each optimization uses a 24 hour horizon and an 8 hour implementation interval. At each iteration, the UC problem is solved over the current optimization horizon. The decisions in the first 8 hour interval are enforced. The optimization horizon then shifts forward by 8 hours. The terminal commitment status and generator outputs provide the initial conditions for the next optimization window.

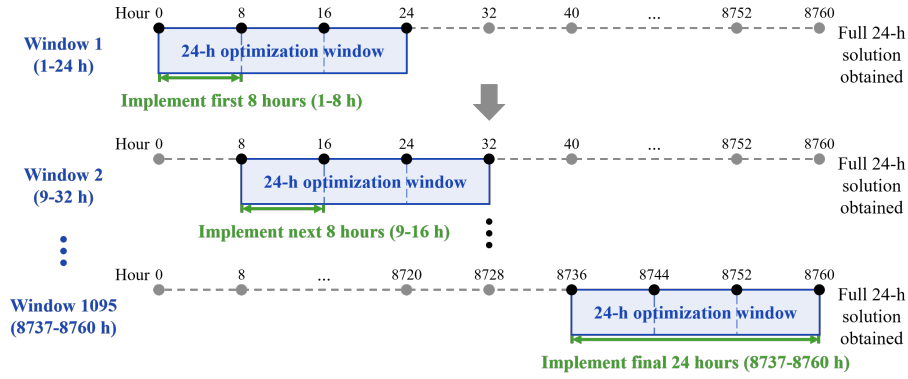


Figure 1: Rolling-horizon scheduling process with a 24 hour optimization window and an 8 hour receding step.

Objective Function

Let $\mathcal{G}$ denote the set of conventional generating units and let $\mathcal{T}$ denote the set of time periods in each optimization window. The rolling-horizon UC model minimizes generation cost, startup cost, renewable curtailment cost, and load shedding cost:

$$\begin{aligned}
 \min \quad & \sum_{g \in \mathcal{G}} \sum_{t \in \mathcal{T}} (C_{\text{op}} P_{g,t} + C_{\text{st},g} u_{g,t}) \\
 & + C_{\text{ren}} \sum_{t \in \mathcal{T}} (P_{\text{wind},t}^{\text{curt}} + P_{\text{pv},t}^{\text{curt}}) \\
 & + C_{\text{load}} \sum_{t \in \mathcal{T}} P_{\text{load},t}^{\text{shed}}
 \end{aligned} \tag{1}$$

Here, C_{op} , $C_{\text{st},g}$, C_{ren} , and C_{load} denote the operating cost coefficient, startup cost coefficient, renewable curtailment penalty, and load shedding penalty, respectively.

Standard UC Constraints

The rolling-horizon UC model includes the following standard constraints.

Additional symbols are defined below. $P_{g,t}$ denotes the dispatched power of unit g at time t . $P_{\text{wind},t}$, $P_{\text{pv},t}$, and $P_{\text{load},t}$ denote the forecast wind power, photovoltaic power, and load power. $P_{\text{wind},t}^{\text{curt}}$ and $P_{\text{pv},t}^{\text{curt}}$ denote the curtailed wind power and photovoltaic power. $P_{\text{load},t}^{\text{shed}}$ denotes the shed load power. P_g^{max} and P_g^{min} denote the maximum and minimum generation limits of unit g . R_t^{min} denotes the minimum spinning reserve requirement. T_g^{up} and T_g^{dn} denote the minimum up and down times of unit g . R_g^{up} and R_g^{dn} denote the ramp up and ramp down rates of unit g . $F_{l,t}$ denotes the active power flow on line l . B_l and x_l denote the line susceptance and reactance. $\theta_{i,t}$ denotes the voltage phase angle at bus i . F_l^{max} denotes the thermal capacity of line l . P_N denotes the system power base. $X_{g,t}$ denotes the binary commitment status of unit g . $u_{g,t}$ and $v_{g,t}$ denote the binary startup and shutdown indicators of unit g .

The power balance constraint matches total supply and served load:

$$\begin{aligned} \sum_g P_{g,t} + P_{\text{wind},t} - P_{\text{wind},t}^{\text{curt}} + P_{\text{pv},t} - P_{\text{pv},t}^{\text{curt}} \\ = P_{\text{load},t} - P_{\text{load},t}^{\text{shed}}, \quad \forall t \end{aligned} \quad (2)$$

The spinning reserve constraint preserves generation headroom:

$$\sum_g (P_g^{\text{max}} X_{g,t} - P_{g,t}) \geq R_t^{\text{min}}, \quad \forall t \quad (3)$$

The minimum up time and down time constraints use standard binary logic:

$$\sum_{\tau=t}^{t+T_g^{\text{up}}-1} X_{g,\tau} \geq T_g^{\text{up}} (X_{g,t} - X_{g,t-1}), \quad \forall g, \forall t \quad (4)$$

$$\sum_{\tau=t}^{t+T_g^{\text{dn}}-1} (1 - X_{g,\tau}) \geq T_g^{\text{dn}} (X_{g,t-1} - X_{g,t}), \quad \forall g, \forall t \quad (5)$$

The ramp rate constraints limit generation changes between consecutive hours:

$$\begin{aligned} P_{g,t} - P_{g,t-1} \leq R_g^{\text{up}} X_{g,t-1} \\ + P_g^{\text{max}} (X_{g,t} - X_{g,t-1}), \quad \forall g, \forall t \end{aligned} \quad (6)$$

$$\begin{aligned} P_{g,t-1} - P_{g,t} \leq R_g^{\text{dn}} X_{g,t} \\ + P_g^{\text{max}} (X_{g,t-1} - X_{g,t}), \quad \forall g, \forall t \end{aligned} \quad (7)$$

The generation limits bound dispatched power for each committed unit:

$$P_g^{\text{min}} X_{g,t} \leq P_{g,t} \leq P_g^{\text{max}} X_{g,t}, \quad \forall g, \forall t \quad (8)$$

The DC power flow equations use the line susceptance $B_l = 1/x_l$:

$$\begin{aligned} F_{l,t} &= B_l(\theta_{i,t} - \theta_{j,t})P_N, \quad \forall l = (i,j), \forall t \\ |F_{l,t}| &\leq F_l^{\max}, \end{aligned} \quad (9)$$

The transition indicators satisfy:

$$u_{g,t} - v_{g,t} = X_{g,t} - X_{g,t-1}, \quad \forall g, \forall t \quad (10)$$

$$u_{g,t} + v_{g,t} \leq 1, \quad \forall g, \forall t \quad (11)$$

The startup cost uses the convex hull constraints $u_{g,t} \geq X_{g,t} - X_{g,t-1}$ and $u_{g,t} \geq 0$. These constraints give the cost term $C_{st,g}u_{g,t}$ in the objective for each feasible binary commitment trajectory.

Embedded RoCoF Constraint

The initial RoCoF after a contingency of magnitude ΔP_{fault} is:

$$\frac{\Delta P_{\text{fault}}}{2 \sum_g M_g X_{g,t}} \leq \text{RoCoF}^{\max} \quad (12)$$

Here, M_g denotes the inertia constant of unit g , ΔP_{fault} denotes the contingency magnitude, and $\text{RoCoF}^{\max}$ denotes the RoCoF limit. This expression is non-linear due to the reciprocal committed inertia term. The RoCoF constraint is reformulated as mixed-integer linear constraints in every optimization window. The constraint maintains sufficient online inertia for the initial post-contingency frequency gradient.

Representative Scenario Screening

The annual UC solution produces distinct commitment patterns. The 15 most frequent patterns are retained. Each pattern is paired with two critical fault locations. This procedure produces 30 planning scenarios. The scenarios provide fixed operating conditions for DSR availability assessment, frequency response simulation, and the two-stage procurement model.